

Dynamic Traffic Control through Intelligent Congestion Detection, Dissemination, and Rerouting

Pranitha Gandham[✉], Megha Vardhan Mirthipati[✉], Vinod Kumar Jain*[✉] *Senior Member, IEEE*, and Shashikala Tapaswi[✉] *Senior Member, IEEE*,

Abstract—Urban traffic congestion continues to be a major challenge in smart cities, primarily due to the ineffective dissemination of congestion alerts and inefficient rerouting of vehicles in traditional Vehicular Ad-hoc Networks (VANETs) based Intelligent Transport Systems (ITS). Existing solutions are primarily based on the use of periodic beaconing or centralized control mechanisms and are generally associated with network overload, high latency, and scalability bottlenecks. This work proposes a dynamic traffic control framework ICDDR (Intelligent Congestion Detection, Dissemination, and Rerouting), that intelligently detects congestion, disseminates real-time updates, and enables effective rerouting to improve the urban traffic scenario. The suggested framework introduces three key innovations: (1) the event-driven data dissemination technique that ensures the delivery of alerts when the congestion threshold is breached and an adaptive TTL mechanism that dynamically controls congestion alert message (CAM) propagation, (2) targeted CAM dissemination through vehicles and Road Side Units (RSUs) during congestion, (3) edge-assisted distributed processing, using Roadside Units (RSUs) as localized computational nodes for efficient routing decisions. This framework has been validated through simulation in a Manhattan Grid scenario and a real-world urban scenario of New Delhi city, taken from Open Street Maps using OMNeT++, Veins, and SUMO. The simulation results demonstrate 32.1% and 29.8% reduced average travel time in Manhattan and New Delhi Urban scenarios, respectively compared to the baselines. ICDDR leads to the lowest CO₂ emissions, its emissions drop by 20.7% in comparison with the baseline models. This model leads to fewer packet collisions, with a reduction of 13.4% in Manhattan and 22.2% in New Delhi urban scenarios, respectively, compared to baselines, while there is a 6.8% decrease in total packet transmission.

Index Terms—Vehicular Ad-hoc Networks (VANETs), Vehicle-to-Vehicle (V2V) Communication, Vehicle-to-Infrastructure (V2I) Communication, Roadside Units (RSUs), Data Dissemination, Traffic Management, Edge Computing, Event-Driven Systems, Intelligent Transportation Systems (ITS).

I. INTRODUCTION

TRAFFIC CONGESTION is an alarming problem on roads worldwide. The increasing population and a change in the mode of transport - usually from walking or public transport to a greater dependence on private vehicles - are the main reasons for severe traffic congestion. The 2023 Urban Mobility Report [1] highlights a growing global traffic problem

- 78% of urban areas experienced increased congestion from 2022 to 2023, with drivers in cities spending an average of 50–60 extra hours in traffic each year. Popular navigation tools like Google Maps [2], Sygic [3] and Waze [4] do offer live traffic updates using crowdsourced data, but they aren't designed for real-time, safety-critical communication between vehicles. Their dependence on mobile internet and cloud services makes them less reliable in areas like tunnels or remote regions, where connectivity is weak or inconsistent.

Vehicular Ad-hoc Networks (VANETs) [5] have become fundamental to modern Intelligent Transportation Systems, enabling critical Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communications. These communications take place through On-Board Units (OBUs) embedded in vehicles and Roadside Units (RSUs) strategically deployed along roads, supporting real-time exchange of traffic data, including congestion notifications, accident warnings, and optimal routing instructions, while addressing challenges in urban mobility management. VANETs play a vital role in smart traffic management by enabling decentralized control, adaptive routing, and predictive traffic flow analytics to mitigate congestion and enhance road safety.

Although VANET-based solutions exist to help solve the problem of traffic in urban areas, they are limited in several ways that prevent them from being effective in a dynamic urban situation. Periodic beaconing leads to the propagation of redundant messages, especially in dense traffic conditions, resulting in inefficient data dissemination. Architectures relying on central servers often suffer from increased latency, scalability bottlenecks, and potential privacy concerns due to their centralized nature. The scalability and efficiency of existing methods are also constrained by the absence of fine-grained message prioritization and proper adaptive behavior to changing traffic. Furthermore, static communication mechanisms are unable to cope with rapidly changing urban traffic dynamics. All these limitations highlight the need to have a distributed, intelligent, and adaptable approach to VANET-enabled traffic management. Thus, this paper proposes an Intelligent Congestion Detection, Dissemination, and Rerouting (ICDDR) architecture for dynamic traffic control using VANET that mainly disseminates congestion alerts by V2I communication along with the localised receptiveness of V2V communication to overcome the above-mentioned limitations of the existing system.

The proposed ICDDR system employs RSUs as edge nodes with computational intelligence to decentralize traffic processing and enable localized congestion monitoring and route optimization. This approach decreases reliance on centralized

Pranitha Gandham is with the Department of Computer Science and Engineering, ABV-IIITM Gwalior. Email: bcs_2022025@iiitm.ac.in

Megha Vardhan Mirthipati is with the Department of Computer Science and Engineering, ABV-IIITM Gwalior. Email: bcs_2022045@iiitm.ac.in

*Corresponding Author, Vinod Kumar Jain is with the Department of Computer Science and Engineering, ABV-IIITM Gwalior. Email: vk-jain@iiitm.ac.in

Shashikala Tapaswi is with the Department of Computer Science and Engineering, ABV-IIITM Gwalior. Email: stapaswi@iiitm.ac.in

infrastructure. The proposed approach also solves the inefficiency of existing systems that rely on periodic triggering or a fixed lifetime of messages, even in smooth traffic conditions, causing unnecessary propagation of data.

The proposed ICDDR framework makes three primary contributions: (1) an event-triggered hybrid communication protocol that replaces fixed-interval beaconing with prioritized messaging, activating updates only when congestion thresholds are breached to significantly reduce redundant transmissions through targeted data dissemination; (2) a distributed processing model that leverages RSU capabilities for edge-level congestion detection and edge weights computation, eliminating reliance on centralized processing and rerouting using A* algorithm; and (3) empirical validation through simulation using OMNeT++ [6], SUMO [7], and Veins [8] and comparing the results to the baseline approaches. The proposed architecture specifically minimizes overhead through dynamic message scheduling and localized computation at RSUs, combining the benefits of infrastructure-assisted coordination with decentralized execution.

The remaining sections of the paper are organized as follows: Section II gives a detailed description of the literature review on congestion detection and data dissemination in VANETs, and rerouting strategies. The Section III is used to discuss the methodology implemented in the paper. Section IV presents the simulation results comparing them with the baseline models. Finally Section V concludes and outlines the possible future work.

II. LITERATURE REVIEW

This section presents an overview of vehicle rerouting systems that employ various strategies to mitigate traffic congestion in VANET. Typically, such systems accomplish two fundamental tasks: real-time detection of traffic congestion, coupled with efficient message dissemination for sharing congestion information in the VANET and implementation of intelligent rerouting mechanisms to circumvent congested areas. These functions are essential for improving traffic flow and reducing travel delays in city environments.

A. Congestion Detection and Data Dissemination

In the literature, various road and traffic parameters have been used for detecting traffic congestion in vehicular networks. Various systems use different metrics to evaluate congestion: DIVERT [9] relies on traffic density (vehicle count relative to road capacity), whereas DEDR [10] primarily focus on the total number of vehicles. On the other hand, De Souza et al. [11] and Rezaei et al. [12] incorporate parameters such as vehicle count, average speed and maximum speed.

Recent research has leveraged deep learning models to improve prediction accuracy. PCNN framework [13] utilized average travel time as input to a deep neural network (DNN), transforming time-series data into 2D matrices and applying convolution layers to extract multi-scale traffic patterns. Additional models like Graph Convolution Neural Networks (GCNNs) [14] and Long Short-Term Memory (LSTM) [13] networks have also shown effectiveness in traffic prediction.

Multiple systems address congestion detection and data dissemination using diverse paradigms. PANDORA [15] applies opportunistic V2V sharing to localize dissemination and reduce overhead, but struggles to adapt due to static intervals and predefined zones. UCONDES [11] uses an artificial neural network to detect congestion through vehicle messages but faces message redundancy when multiple vehicles report the same incident.

Fully distributed solutions like FIRE-NRD [16] enable individual vehicles to detect and broadcast congestion independently. Although scalable and infrastructure-free, they lack a global view, limiting optimal routing. Hybrid approaches, such as the fog-cloud-based rerouting system proposed by Tseng and Ferng [17], adapt road weights dynamically, reducing travel time by up to 20% compared to static models. Conversely, V2V-only solutions like RIDER [18] implement adaptive Time-To-Live (TTL) to limit alert propagation without infrastructure dependence. While this reduces centralization, it performs poorly in high-density traffic due to fixed TTL settings, risking broadcast storms and computational overhead.

B. Vehicle Rerouting

Once traffic congestion is detected, the key challenge is figuring out how to reroute vehicles efficiently. The simplest solution is to use Dijkstra's algorithm [19] to calculate the shortest alternate path. However, if every vehicle uses the same shortest path, it can lead to a new congestion hotspot. To solve this, researchers are now focusing on more balanced and cooperative rerouting strategies.

A promising alternative is the k-shortest path (kSP) [20] algorithm, which offers multiple loop-free path options instead of just one. The choice of path depends on how edge weights are assigned - these might reflect traffic speed, travel time, or vehicle count. For example, DIVERT [9] focus solely on travel time. Tseng et al. [17] enhance this by including road capacity, traffic density, and speed, enabling more precise and real-time route planning.

Since kSP [20] can be computationally expensive in large networks, it's often used in interval-based rerouting, where only a subset of vehicles are rerouted at fixed times. However, selecting which vehicles to reroute is not straightforward. Some systems rely on hop count - the number of turns a vehicle is from the congestion point-but this method can misjudge actual distances and reroute vehicles unnecessarily.

To address this, Ho et al. [21] proposed a dynamic rerouting model that considers how close a vehicle is to the congested area and the available capacity of nearby roads. This improves the efficiency of rerouting and reduces unnecessary detours.

Rerouting strategies also differ in timing. Some are reactive-responding to current congestion, while others are proactive, anticipating congestion before it happens. RIDER [18], for example, combines historical and real-time data to make better decisions at the system level. Silva [22] takes a centralized approach that computes two route options for each vehicle and selects the best one based on real-time road conditions. However, centralized systems often struggle to scale in dense traffic due to high computational demands. Yusuke et al. [23]

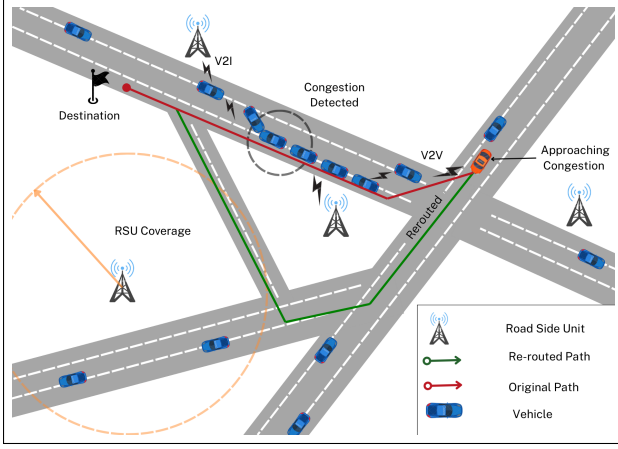


Fig. 1: Representation of the Proposed System Architecture

introduced a flow-based centralized routing framework (FCR) that explicitly incorporates segment capacity into detour path prioritization, achieving near-optimal traffic distribution.

Despite advancements in vehicle rerouting and congestion control, existing methods still face key challenges. Many [15]- [18], [23] rely on static or message broadcasting, which leads to unnecessary redundancy and higher network load. In contrast, our system uses relevance-based rebroadcasting, allowing only the most appropriate vehicles to forward congestion alerts. Unlike traditional fixed-interval beaconing, we adopt an event-triggered TTL mechanism that adjusts message propagation in real-time based on congestion severity.

Moreover, while few frameworks combine live traffic data with adaptive routing, our approach introduces a congestion-aware k-shortest A* algorithm that dynamically recalculates routes. To strike a balance between communication efficiency and responsiveness, we introduce a Suitability Index that factors in directional relevance and communication distance, ensuring efficient and timely dissemination.

Overall, the proposed Intelligent Congestion Detection, Dissemination, and Rerouting (ICDDR) framework combines V2V and V2I communication with RSU-driven decisions to enable scalable, real-time rerouting with minimal network overhead in urban traffic scenarios [24].

III. METHODOLOGY

This work proposes a novel congestion-aware vehicular communication framework, ICDDR, that ensures reliable data dissemination and intelligent vehicle rerouting in urban environments.

A. System Model and Assumptions

In VANET, all vehicles are equipped with a global positioning system (GPS), an onboard unit (OBU), and are capable of communicating with other vehicles and Road Side Units (RSUs) using IEEE 802.11p interface through Vehicle to Vehicle (V2V) and Vehicle to Infrastructure (V2I) communications. Fig. 1 represents the overall system design and interaction among vehicles and Road Side Units (RSUs). This work makes the following general assumptions:

- Every vehicle is equipped with a map of the city's road network.
- The range of the IEEE802.11p interface is 500 meters.
- RSUs are deployed at 1000-meter intervals, forming the backbone of the network, with vehicles connecting to them directly through V2I links wherever possible.
- In areas where RSU coverage isn't available, vehicles automatically switch to multi-hop V2V communication using up to 4 hops (each hop covering a maximum distance of 500 meters) to maintain continuous connectivity.

With all these features built, every car can process, evaluate, communicate, and make scattered decisions.

The VANET environment can be represented by a directed graph $G = (V, E)$, where V denotes the intersections of the roads and is represented as the set $V = \{v_a, v_b, v_c, \dots\}$. The road segments that connect the road intersections v_i and v_j are represented by edges $e_{ij} = (v_i, v_j)$, where the road segments are denoted by the set $E, \forall (v_i, v_j)$. The traveling cost for the segment e_{ij} is represented by the cost w_{ij} . The set of weights is given by $W = \{w_{ij} \mid i \neq j\}$, where $w_{ij} \in \mathbb{R}_+^*$.

The set $N = \{n_a, n_b, n_c, \dots, n_m\}$ represents the vehicles, where m is the number of vehicles in the road network. Each vehicle and RSU maintain a record called the Road Segment Database, denoted by Ψ , which stores traffic information either observed directly or received from others.

The road capacity of a segment e_{ij} , denoted by $C(i, j)$, is calculated as:

$$C(i, j) = \frac{L(i, j)}{V_{\text{len}}^{\text{avg}} + g_{\text{min}}} \quad (1)$$

where $L(i, j)$ is the length of the road segment, $V_{\text{len}}^{\text{avg}}$ is the average length of a vehicle (5 meters [25]), and g_{min} is the minimum gap (2 meters [26]) between vehicles.

B. Congestion Detection

The proposed system detects traffic congestion based on two primary parameters. The first is the congestion factor α , defined as the ratio of a vehicle's average speed over the past 20 seconds (v_{20}) to the free-flow speed ($\bar{v}_{ij}$) of the corresponding road segment, retrieved from the Road Segment Database Ψ . This metric indicates how closely the current traffic conditions align with ideal, non-congested flow.

When the congestion factor α drops below 0.3, it implies that the vehicle's speed has reduced by 70% or more compared to the free-flow speed. This threshold is based on empirical studies and aligns with established congestion detection methods. [18] This suggests that the vehicle is moving considerably slower than the ideal speed for that road segment, indicating a high likelihood of congestion. Hence, the first condition for congestion is fulfilled when:

$$\alpha \leq 0.3 \quad \text{where} \quad \alpha = \frac{v_{20}}{\bar{v}_{ij}} \quad (2)$$

Although the average speed on congested roads tends to be low and higher on free-flowing roads, a vehicle moving at a lower speed does not necessarily imply that the road segment is congested. Therefore, we also consider another important

Algorithm 1 FindCongestion

Require: Congestion factor α , threshold α_{th} , weight factor λ , congestion threshold CD_{th}

//Compute congestion factor

- 1: $\alpha = \frac{v_{20}}{v_{ij}}$ //Performed at the OBUs in vehicles.
- 2: **if** $\alpha \leq \alpha_{th}$ **then**
- 3: Vehicle sends alert message to nearby RSUs
- 4: **end if**
- // RSUs aggregate received α values
- 5: $C(i, j) = \frac{L(i, j)}{V_{len}^{avg} + g_{min}}$ //Compute segment capacity
- 6: $D(i, j) = \frac{N_v(i, j)}{C(i, j)}$ // Compute road density
- 7: $CD = \lambda(1 - \alpha) + (1 - \lambda) \cdot D(i, j)$ //Compute congestion score
- 8: **if** $CD > CD_{th}$ **then**
- 9: Mark segment (i, j) as congested
- 10: RSU broadcasts congestion alert message
- 11: Initiate rerouting procedure using updated weights
- 12: **end if**

parameter—road density $D(i, j)$, which reflects the number of vehicles present within a given road segment.

The density of the current segment is monitored by RSUs and is defined as the ratio between the number of vehicles $N_v(i, j)$ on the current segment (i, j) and the maximum capacity of the road segment.

$$D(i, j) = \frac{N_v(i, j)}{C(i, j)}, \quad \text{where} \quad C(i, j) = \frac{L(i, j)}{V_{len}^{avg} + g_{min}} \quad (3)$$

Once the congestion threshold based on the first factor is exceeded, the vehicle sends an alert message to nearby RSUs. The RSUs then aggregate the received α values and combine them with the computed road density $D(i, j)$ for that segment to assess the level of congestion more accurately.

The RSUs compute a weighted congestion score CD using the following formula:

$$CD = \lambda(1 - \alpha) + (1 - \lambda) \cdot D(i, j) \quad (4)$$

where λ determines the relative importance assigned to speed (α) and vehicle density, with its value established as 0.1 [17]. If the resulting score exceeds a predefined threshold, the road segment is classified as congested, triggering the dissemination of alert messages and the initiation of rerouting procedures. The process of detecting congestion and disseminating alerts is outlined in Algorithm 1. In the following subsection, the data dissemination based on the relevance of the vehicles is discussed.

C. Relevance based Data Dissemination

The RSUs create a Congestion Alert Message (CAM) that includes the segment ID, direction, and severity of the congestion. These messages are disseminated to the nearest vehicles using Vehicle-to-Infrastructure (V2I) mode of communication.

Each CAM is assigned an adaptive Time-To-Live (TTL) value. TTL represents the maximum number of times a

message can be forwarded before it is discarded. This adaptive mechanism allows efficient utilization of network resources—using sparse message propagation in free-flow conditions and more aggressive dissemination in congested scenarios.

The TTL value is dynamically adjusted based on the congestion score CD as follows:

$$TTL = \begin{cases} 4, & CD \geq 0.7 \\ 3, & 0.4 < CD < 0.7 \\ 2, & 0.2 < CD \leq 0.4 \\ 1, & 0.1 < CD \leq 0.2 \end{cases} \quad (5)$$

where TTL denotes the Time-To-Live of the transmitted message and CD is the congestion factor. By appropriately adjusting TTL values, the system effectively minimizes redundant message propagation, thereby reducing unnecessary network flooding and improving overall communication efficiency. The RSUs transmit the CAM message to all the nearest vehicles using the V2I mode of communication. Like in all the baseline models, the vehicles upon receiving CAM, instead of flooding the messages to all the nearby vehicles, we find the most relevant vehicle, and they disseminate the CAM message.

For estimating the relevant vehicles [24], each vehicle continuously maintains a record of its last five GPS positions. Upon receiving a Congestion Alert Message (CAM), the vehicle computes its Travel Path Line (TPL) using the Least Squares method [27].

The relevance R of a vehicle is determined based on its likelihood of approaching the sender vehicle that transmitted the most recent Congestion Alert Message (CAM). The value of R ranges between 0 and 1 and is computed using the slope and Y-intercept of the Travel Path Lines (TPLs) of both the sender and the receiver.

First, the acute angle θ between the TPLs of the sender and receiver is calculated using the following equation:

$$\theta = \tan^{-1} \left(\frac{M_s - M_r}{1 + M_s \cdot M_r} \right) \quad (6)$$

where M_s and M_r are the slopes (tangents) of the sender's and receiver's TPLs, respectively. A small value of θ indicates that both vehicles are moving in similar directions, possibly on the same road.

However, if the two vehicles are on different but nearly parallel roads, the angle θ might still be small, but the difference in their Y-intercepts ΔC will be significant. Thus, R is also dependent on $\Delta C = |C_s - C_r|$, where C_s and C_r are the Y-intercepts of the sender and receiver. Considering both θ and ΔC , the relevance R is computed using Algorithm 2. The value of R is adjusted as follows:

$$R(\theta, \Delta C) = \begin{cases} 1, & \text{if } \theta < \theta_t \text{ and } \Delta C < C_t \\ 0.5 + 0.5 \cdot \frac{\theta}{90}, & \text{if } \theta \geq \theta_t \text{ and vehicle is} \\ & \text{moving toward sender} \\ 0.5 \cdot \frac{\theta}{90}, & \text{if } \theta \geq \theta_t \text{ and vehicle is} \\ & \text{moving away from sender} \end{cases} \quad (7)$$

Upon calculating the relevance score R , each vehicle independently computes another metric called the *Suitability Index* (S_i), which quantifies how appropriate the vehicle is to act as the next-hop rebroadcast node for a Congestion Alert Message (CAM). The suitability index S_i is computed as:

$$S_i = \rho \cdot R + \gamma \cdot D_{\text{norm}} \quad (8)$$

where $R \in [0, 1]$ is the relevance score, and $D_{\text{norm}} \in [0, 1]$ is the normalized distance between the sender and the receiver vehicles within the communication radius R_c . The normalized distance is given by:

$$D_{\text{norm}} = \frac{D_O}{R_c} \quad (9)$$

Here, D_O is the actual distance between the sender and the receiver, and R_c is the communication range of the sender.

To balance between the relevance of message dissemination and the speed of propagation, we introduce two dynamic tuning weights ρ and γ , subject to the constraint $\rho + \gamma = 1$. Assigning a higher value to ρ prioritizes dissemination relevance by selecting vehicles that are directionally aligned, though it may slow down the process as nearby vehicles are repeatedly selected. Conversely, prioritizing γ favors faster dissemination by selecting farther vehicles, even if they are less directionally aligned.

To maintain a balanced trade-off between relevance and dissemination speed, our system dynamically updates the values of ρ and γ based on three contextual factors: (1) Urgency level of the message, (2) Local traffic density, (3) Time of day. The values of ρ and γ are estimated using the procedure described in Algorithm 3.

Once the values of ρ and γ are computed, all vehicles that receive the CAM evaluate their own suitability index. The vehicle with the highest S_i value is considered the most appropriate to forward the CAM, thereby ensuring relevance-aware yet timely dissemination. This architecture guarantees that vehicles entering congested regions receive up-to-date routing information. The next subsection discusses the initiation of the weight update protocol within the vehicular environment.

D. Dynamic Weighted Graph Update

To enable real-time adaptability in routing, this work proposes a dynamic graph weight updating mechanism using Road Side Units (RSUs). Upon detecting congestion alerts, RSUs exchange local traffic parameters such as average speed $\bar{v}_{ij}$, vehicle density $N_v(i, j)$, and segment capacity $C(i, j)$. Segment weights are computed using:

$$w_{ij}(t) = \frac{L(i, j)}{\bar{v}_{ij}} \cdot \left(1 + \frac{N_v(i, j)}{C(i, j)}\right)^2 \quad (10)$$

This formulation accounts for segment length, congestion, and average speed, thereby penalizing heavily loaded or slow routes. To conserve bandwidth and avoid redundant broadcasts, updates are triggered only when the relative change exceeds a threshold w_{th} :

Algorithm 2 Relevance Calculation of Receiver

Require: Sender's slope M_s , Y-intercept C_s , position $P_s(x_s, y_s)$;
 1: Receiver's slope M_r , Y-intercept C_r , position $P_r(x_r, y_r)$, fifth last position $P_{r5}(x_{r5}, y_{r5})$
Ensure: Relevance R of the receiver
 2: Initialize threshold values θ_t , C_t
 // Compute acute angle between sender and receiver TPLs using [6]
 3: **if** $\theta > 90$ **then**
 4: $\theta = 180 - \theta$
 5: **end if**
 6: $\Delta C = |C_s - C_r|$ //Compute intercept difference
 7: **if** $\theta < \theta_t$ **and** $\Delta C < C_t$ **then**
 8: $R = 1$
 9: **else**
 10: $D_0 = \sqrt{(x_s - x_r)^2 + (y_s - y_r)^2}$
 //Compute current distance from sender
 11: $D_5 = \sqrt{(x_s - x_{r5})^2 + (y_s - y_{r5})^2}$
 //Compute distance from fifth last position:
 12: **if** $D_0 < D_5$ **then** // Receiver is moving toward sender
 13: $R = 0.5 + 0.5 \cdot \frac{\theta}{90}$
 14: **else**
 15: $R = 0.5 \cdot \frac{\theta}{90}$
 16: **end if**
 17: **end if**

Algorithm 3 Suitability Index Calculation Based on Context-Aware Weights

Require: Urgency level $U \in [0, 1]$;
 1: Traffic density $D \in [0, 1]$;
 2: Time factor $T \in [0, 1]$;
 3: Relevance score $R \in [0, 1]$;
 4: Normalized distance from sender $D_{\text{norm}} \in [0, 1]$
 5: Define dynamic tuning weights: $w_1 = 0.5$, $w_2 = 0.3$, $w_3 = 0.2$
 6: $\rho = w_1 \cdot U + w_2 \cdot (1 - D) + w_3 \cdot T$
 //weight scaling factor
 7: **if** $\rho > 0.9$ **then** // Clamping ρ for stability
 8: $\rho = 0.9$
 9: **else if** $\rho < 0.5$ **then**
 10: $\rho = 0.5$
 11: **end if**
 12: $\gamma = 1.0 - \rho$ //Complementary weight
 13: $S_i = \rho \cdot R + \gamma \cdot D_{\text{norm}}$
 //Suitability Index
 14: **return** S_i

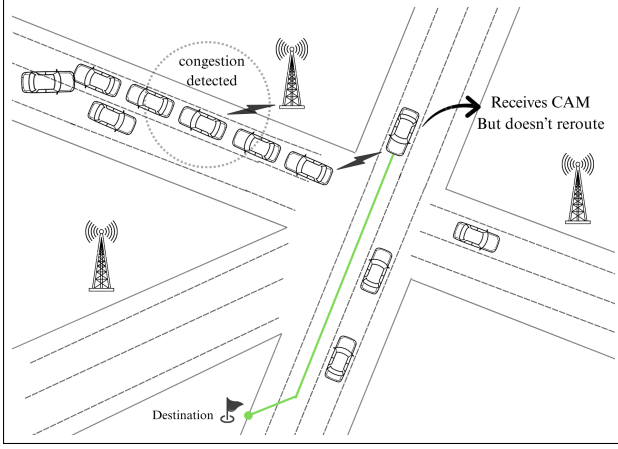


Fig. 2: Rerouting Scenario

$$\frac{|w_{ij}^{\text{old}} - w_{ij}^{\text{new}}|}{w_{ij}^{\text{old}}} > w_{th} \quad (11)$$

These updated weights are synchronized across RSUs to maintain consistency in the network view. Vehicles then recompute optimal paths using a modified A* algorithm based on the new dynamic weights.

E. Vehicle Re-routing

To enable responsive and scalable rerouting in dynamic vehicular networks, this work employs a congestion-aware extension of the classical A* algorithm, formalized in Algorithm 4. Upon reception of a Congestion Awareness Message (CAM) from Roadside Units (RSUs) or proximate vehicles, the receiving node verifies whether its current path to the destination intersects with any known congested edges. If such segments are present, a rerouting procedure is initiated; otherwise, the original path is retained, thereby avoiding unnecessary recomputation.

This algorithm modifies the standard A* search to support the computation of multiple viable alternatives by incorporating three key enhancements: (1) Nodes may be revisited via distinct paths to explore diverse routing options. (2) The search does not terminate upon first reaching the destination, but continues until k loopless, cost-optimal paths are identified. (3) In scenarios where fewer than k complete paths are discovered, the algorithm returns the most promising partial paths observed, enabling continued navigation under high congestion.

The rerouting procedure is encapsulated in two primary functions. The `RerouteIfNeeded` function iterates over each edge in the current path to check for congestion. If any edge lies within the set of congested segments C , the vehicle invokes `KShortestPaths` to compute alternative routes. As shown in Fig. 2, although the vehicle is provided with a CAM, it will not be rerouted since there is no edge in the route in which the vehicle is taking which can take the vehicle to the congested road segments.

The `KShortestPaths` function operates using a priority queue to maintain candidate paths sorted by their f -score (sum

Algorithm 4 Congestion-Aware k -Shortest A* Rerouting Algorithm

Require: Graph $G = (V, E)$, source s , destination d , number of paths k , congested edge set C , vehicle's current path P_{curr} , coordinates $pos[v] = (x_v, y_v)$

Ensure: Top- k shortest paths from s to d if congestion in current path

```

1: function REROUTEIfNEEDED( $P_{curr}, C$ )
2:   for each edge  $e$  in  $P_{curr}$  do
3:     if  $e \in C$  then
4:       return KSHORTESTPATHS( $s, d, k, G, C$ )
5:     end if
6:   end for
7:   return  $P_{curr}$  // No rerouting needed
8: end function
9: function KSHORTESTPATHS( $s, d, k, G, C$ )
10: // Initialize priority queue  $Q$  to store paths with  $f$ -score
11: // Initialize list  $KPaths = []$  to store  $k$  best paths
12: Define  $h(v) = \sqrt{(x_v - x_d)^2 + (y_v - y_d)^2}$ 
13: Insert initial path  $[s]$  with  $g = 0, f = h(s)$  into  $Q$ 
14: while  $Q \neq \emptyset$  and  $|KPaths| < k$  do
15:   Extract path  $P$  with lowest  $f$  from  $Q$ 
16:   Let  $u = \text{last node in } P$ 
17:   Compute  $g(P)$  as cumulative cost of  $P$ 
18:   if  $u = d$  then
19:     Add  $P$  to  $KPaths$ 
20:     continue
21:   end if
22:   for each neighbor  $v$  of  $u$  do
23:      $P' = P$  extended with  $v$ 
24:      $g(P') = g(P) + w_{uv}(t)$ 
25:      $f(P') = g(P') + h(v)$ 
26:     Insert  $P'$  into  $Q$  with priority  $f(P')$ 
27:   end for
28: end while
29: if  $|KPaths| = 0$  then
30:   return best  $k$  paths seen so far from  $Q$ 
31: else
32:   return  $KPaths$ 
33: end if
34: end function

```

of actual cost and heuristic estimate). The heuristic function $h(v)$ is defined as the Euclidean distance between the current node v and the destination d :

$$h(v) = \sqrt{(x_v - x_d)^2 + (y_v - y_d)^2} \quad (12)$$

The algorithm initializes with a trivial path consisting of the source node s , and iteratively extends it by evaluating all reachable neighbors. For each extension, it calculates the cumulative path cost and updates the queue. Completed paths reaching the destination are stored until k such routes are identified or the search space is exhausted. In the latter case, the best available partial routes are returned to ensure continuity of navigation.

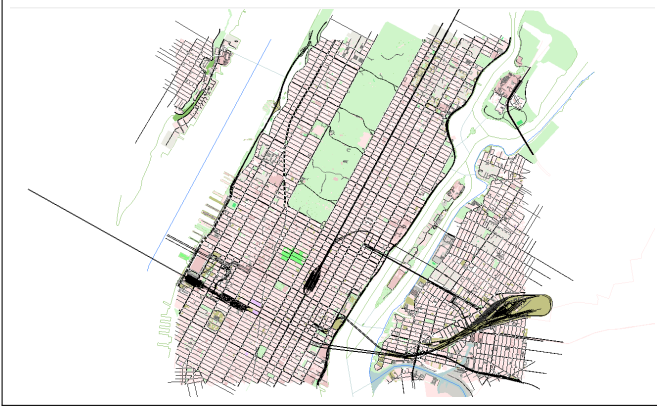


Fig. 3: Map of the Region of Interest in Manhattan Scenario

This hybrid strategy, combining congestion-awareness with a k -shortest path formulation, proves particularly effective in urban VANET contexts where localized and frequently updated traffic information is essential. When integrated with RSU-assisted dynamic edge weight updates, the proposed approach supports sub-second path recalculation while maintaining robustness against mid-route congestion. The algorithm further enhances fault tolerance by ensuring route alternatives are precomputed, mitigating the impact of sudden traffic events. The efficacy of this method is validated in simulation scenarios detailed in the subsequent section.

TABLE I: Simulation Parameters

Parameter	Value
Simulation Area	Manhattan Grid (6.25km ²), New Delhi OSM (10km ²)
Vehicle Density	50-250 veh/km ²
RSU Coverage	1000m radius
Transmission Power	20mW
Bitrate	6Mbps
Frequency	5.89GHz
Propagation Model	Obstacle Shadowing
Fading Model	Jakes' Rayleigh
Shadowing Model	Log-normal ($\sigma=3$ dB)
Antenna Model	Monopole (1.895m height)
MAC Protocol	IEEE 802.11p EDCA
PHY Model	IEEE 1609.4
Beacon Interval	Adaptive (0.2-5Hz)
Max V2V Hops	4
V2V Range	500m
Path Loss Model	WINNER+ B1
Noise Floor	-98dBm
Simulation Time	1600s
SUMO Timestep	180ms

IV. RESULTS AND DISCUSSION

A. Simulation Setup

The proposed system was evaluated through simulations conducted using OMNeT++ 6.1 [6], SUMO 1.22.0 [7], and Veins 5.2 [8]. Two simulation environments were employed: a synthetic 2.5 km $\times$ 2.5 km Manhattan grid (Fig. 3) for controlled testing and a 4 km $\times$ 2.5 km (Fig. 4) real-world urban scenario based on the New Delhi city road network, extracted from OpenStreetMap.

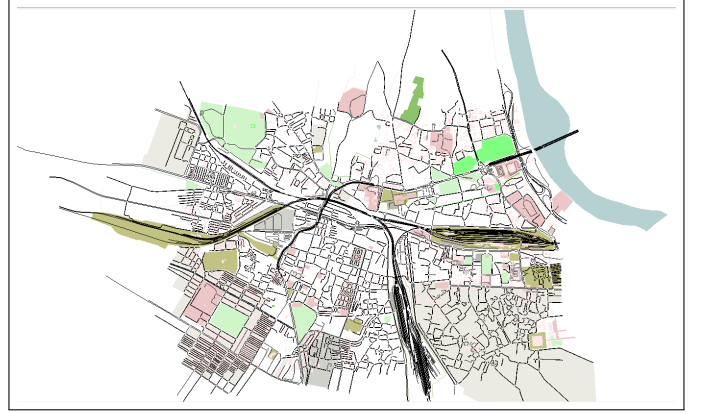


Fig. 4: Map of the Region of Interest in New Delhi Scenerio

As outlined in Table I, the simulations used the IEEE 802.11p EDCA MAC protocol operating at 5.89 GHz with a shadowing propagation model to accurately reflect realistic wireless communication conditions in urban vehicular networks. Vehicle-to-vehicle (V2V) communication was restricted to a maximum of four hops, and adaptive beaconing intervals ranging from 0.2 Hz to 5 Hz were implemented to support scalable information dissemination.

To ensure both reproducibility and practical relevance, the simulations were conducted across a range of traffic densities, varying from 50 to 250 vehicles/km². Randomized accident events and speed fluctuations were introduced to emulate dynamic traffic behaviors. Each simulation ran for 1600 seconds with a 1-second update interval, during which key performance metrics—average travel time, lost time, transmitted messages, packet collisions, number of vehicles rerouted—were recorded.

The proposed ICDDR system was benchmarked against existing solutions including PANDORA [15], RIDER [18] and, FCR [23] allowing for a thorough assessment of its performance and applicability in both structured and unstructured urban environments.

B. Selection of Parameters

1) *Selection of Parameters (w_1, w_2, w_3):* In this study, the dynamic tuning weights are set to $w_1 = 0.5$, $w_2 = 0.3$, and $w_3 = 0.2$. According to this configuration, the greatest importance is assigned to the *Urgency of the message* (w_1), followed by the inverse of *Traffic Density* (w_2), and finally the general *Travel Time* (w_3). These values were identified as yielding the most stable and optimal performance in simulations, based on extensive preliminary trials and empirical evaluations across multiple scenarios. The statistical data are presented in Fig. 5

While dynamic or adaptive weighting mechanisms represent a promising direction for future research, the fixed-weight scheme adopted here provides a robust and consistent foundation. This allows for an objective evaluation of the system's primary contributions without the added complexity of adaptive tuning. Therefore, the selected weights can be regarded as the optimal configuration within the scope of this study.

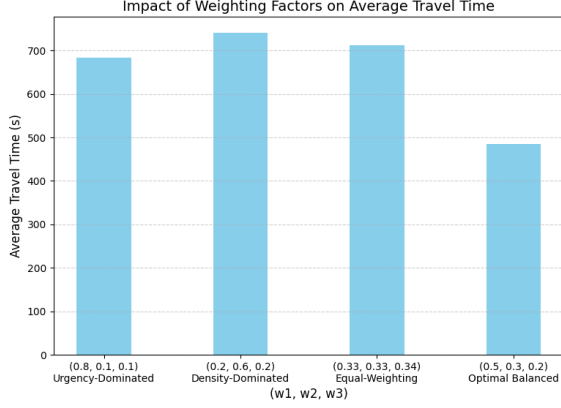


Fig. 5: Travel Times under different sets of (w_1, w_2, w_3)

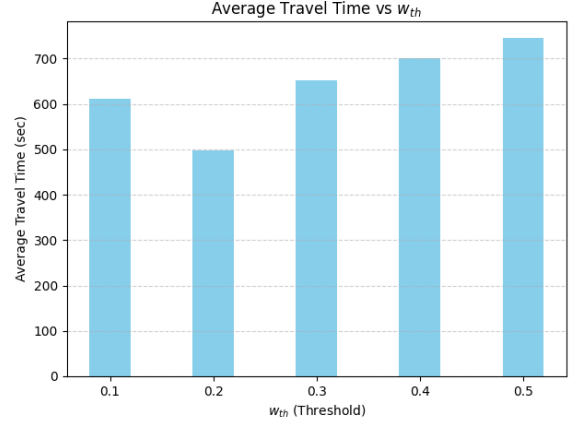


Fig. 6: Travel Time vs parameter w_{th}

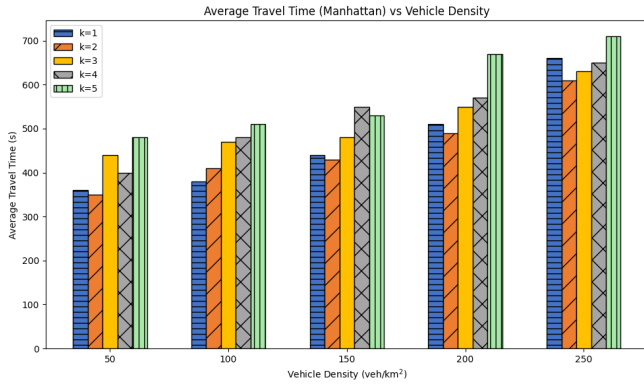


Fig. 7: Result of k in Manhattan Scenario

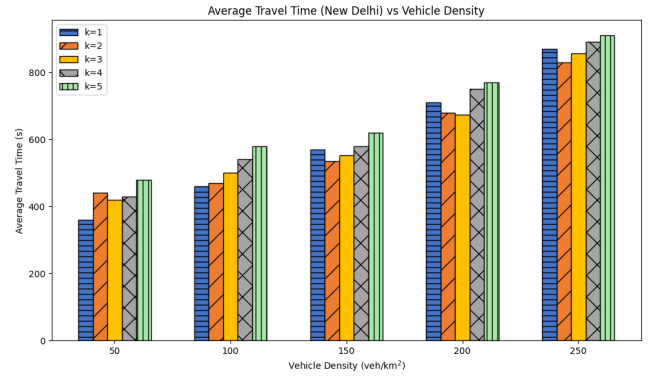


Fig. 8: Result of k in New Delhi Scenario

2) *Determination of w_{th}* : The weight update threshold is comparatively analyzed in Fig.6

The results indicate that a moderate threshold value of $w_{th} = 0.2$ achieves the best performance, yielding the lowest average vehicle travel time.

When the threshold is set too low (i.e., highly sensitive), the travel time increases significantly. This degradation arises from excessive network overhead, as frequent broadcasts caused by minor fluctuations in traffic load congest the communication channel and delay the dissemination of critical information. Conversely, when the threshold is set too high (i.e., less sensitive), the system becomes sluggish in adapting to changing traffic conditions. Vehicles then rely on outdated routing information, leading to inefficient path selection and increased delays.

Therefore, the analysis demonstrates that $w_{th} = 0.2$ represents an optimal balance: it minimizes average travel time while maintaining both communication efficiency and responsiveness to dynamic traffic conditions.

3) *Estimating k* : To ensure that our proposed ICDDR framework adapts effectively to diverse urban traffic conditions, a series of simulation-based experiments were conducted to estimate the key operational parameter k . In our context, k denotes the number of alternative paths computed in the k -A* path finding algorithm, which directly influences the routing

flexibility and adaptability of the system.

We evaluated the performance of the framework for k values ranging from 1 to 5. These simulations were designed to assess key metrics such as average travel time, network throughput, and congestion reduction under varying traffic densities and dynamic conditions.

The results, as depicted in Fig. 7 and Fig. 8, revealed that while $k=1$ and $k=2$ offered limited adaptability due to fewer routing options, increasing k beyond 3 led to a marginal gain in path diversity at the cost of increased computational overhead and route instability. Among the tested values, $k=3$ consistently yielded the most balanced performance achieving significant improvements in traffic flow efficiency without introducing excessive complexity or instability.

Therefore, based on empirical observations, we selected $k=3$ as the optimal value for the ICDDR framework in all subsequent simulations and evaluations.

C. Analysis and Discussion

The experimental evaluation clearly demonstrates the effectiveness of the ICDDR framework, showcasing substantial improvements across multiple performance dimensions—travel time efficiency, no of reroutes, network overhead, and CO₂ emissions—in both the structured Manhattan grid and the more complex, heterogeneous layout of New Delhi city.

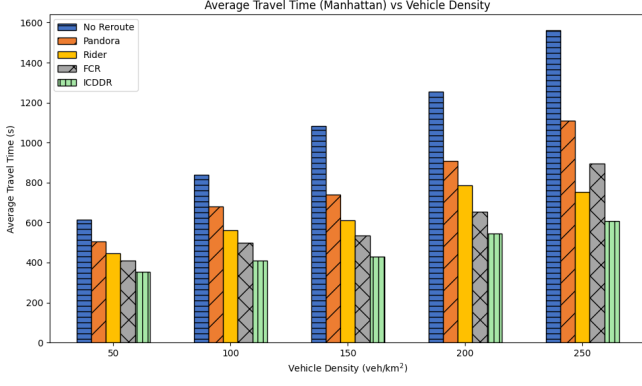


Fig. 9: Average Travel Time- Manhattan Scenario

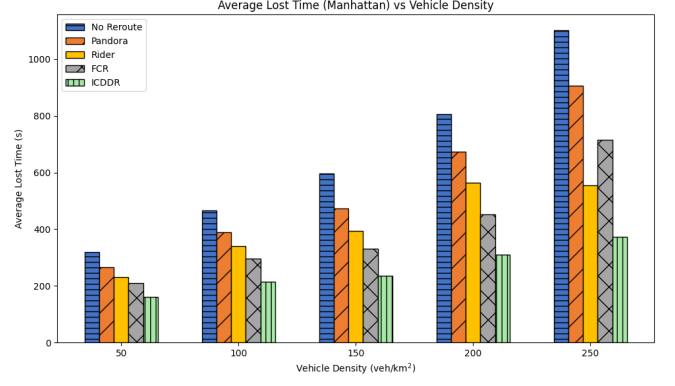


Fig. 10: Average Lost Travel Time- Manhattan Scenario

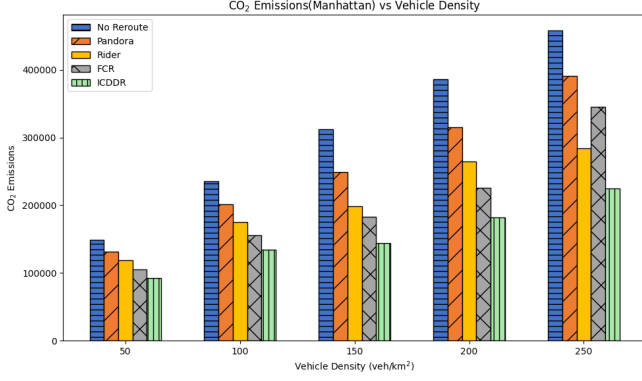


Fig. 11: Total CO2 Emission- Manhattan Scenario

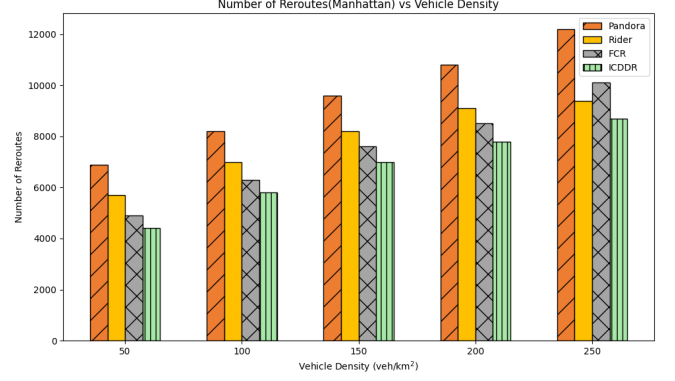


Fig. 12: Total No of Reroutes- Manhattan Scenario

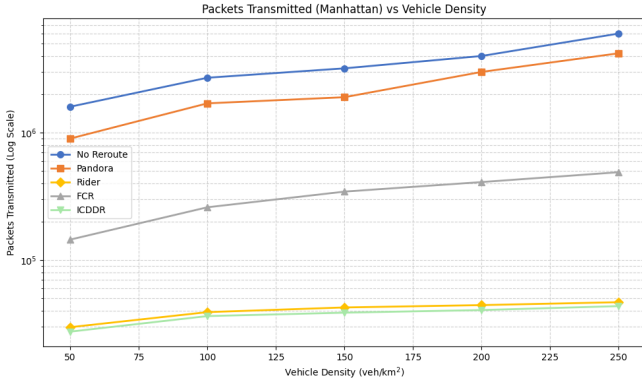


Fig. 13: Total Packets Transmitted- Manhattan Scenario

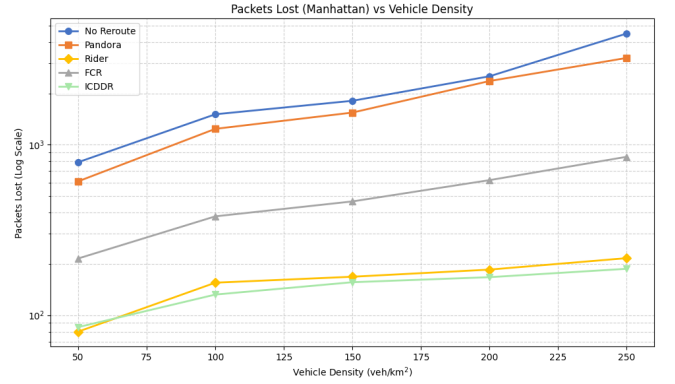


Fig. 14: Total Packets Lost- Manhattan Scenario

As depicted in Fig. 9 and Fig. 15, ICDDR consistently achieves the lowest average travel time, particularly under high-density traffic conditions. For example, at a density of 250 vehicles/km² in Manhattan, ICDDR reduces average travel time by 32.1% compared to FCR, 19.3% compared to RIDER, and a significant 45.2% compared to PANDORA. A similar trend is observed in New Delhi, with reductions of 29.8%, 18.3%, and 43.1%, respectively. These gains are also reflected in the Lost Time metric (see Fig. 10 and 16), where ICDDR significantly reduces time spent in congestion.

These improvements stem from ICDDR's intelligent, event-triggered dissemination architecture. Unlike PANDORA, which uses fixed-interval beaconing, ICDDR triggers communication only when congestion thresholds are breached,

leading to faster and more relevant responses. In contrast, FCR's interval-based centralized routing introduces latency, and RIDER, though adaptive and efficient locally, lacks a global traffic view. ICDDR overcomes these limitations by employing RSUs as edge nodes, dynamically updating network graph weights and routing paths using a modified A* algorithm. In most of the scenarios, the reroute levels were consistently reduced when using routing methods over the baseline, with the ICDDR being the least reroutes on average followed by RIDER and FCR, whereas PANDORA routinely recorded relatively higher reroutes than the best results (Fig. 12 and 18).

ICDDR also demonstrates superior network efficiency, as shown in Fig. 13 and 19. At high density, ICDDR reduces

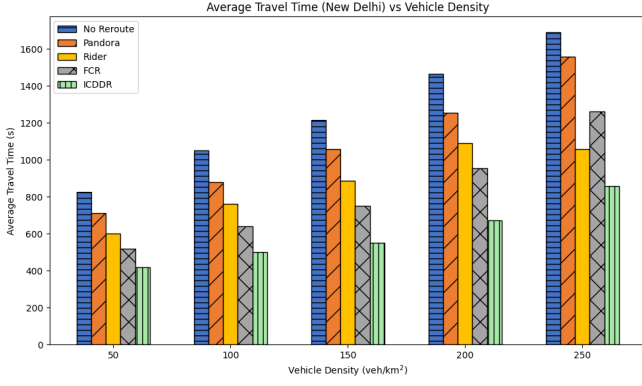


Fig. 15: Average Travel Time- New Delhi Scenario

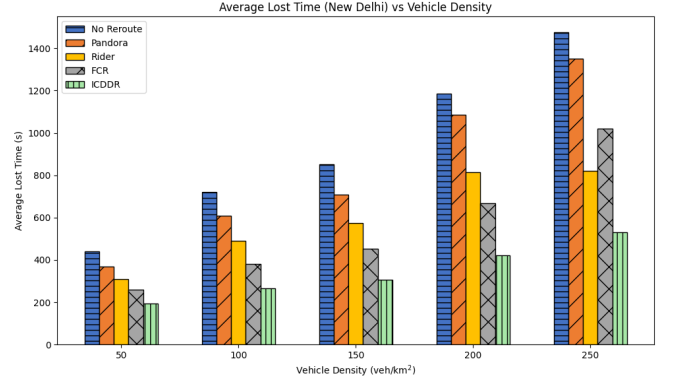


Fig. 16: Lost Travel Time- New Delhi Scenario

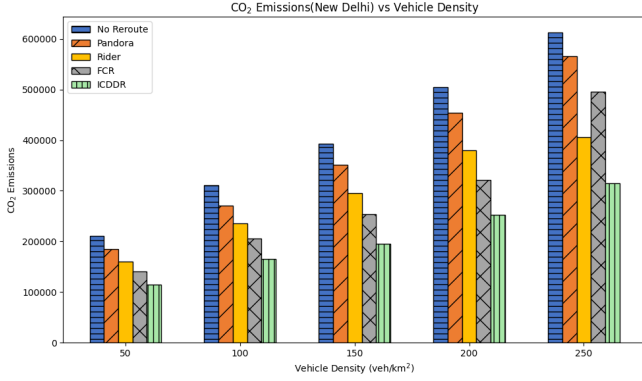


Fig. 17: Total CO2 Emission- New Delhi Scenario

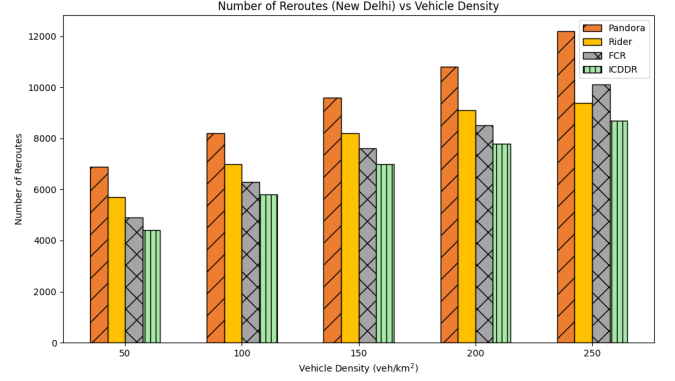


Fig. 18: Total No of Reroutes- New Delhi Scenario

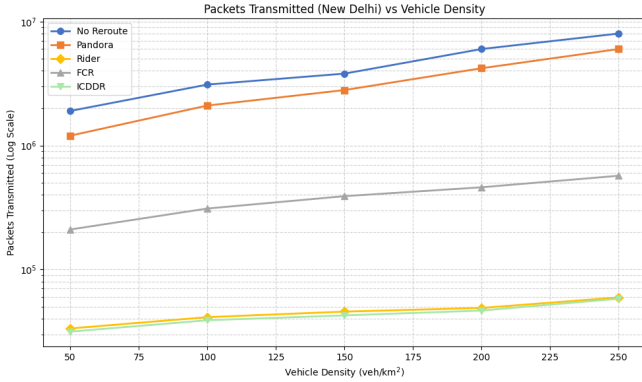


Fig. 19: Total Packets Transmitted- New Delhi Scenario

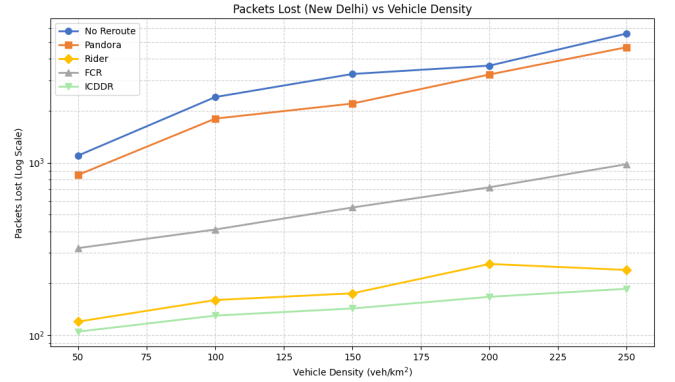


Fig. 20: Total Packets Lost- New Delhi Scenario

total message transmissions by 91.1% versus FCR and 99.0% compared to PANDORA in Manhattan. Even compared to the low-overhead RIDER, ICDDR transmits 6.8% fewer messages. This efficiency leads to fewer packet collisions, with reductions of 13.4% in Manhattan (Fig. 14) and 22.2% in New Delhi scenario (Fig. 20).

These gains are due to ICDDR's relevance-based dissemination strategy. Rather than flooding the network like PANDORA or relying on directional forwarding like RIDER, ICDDR selects only the most relevant vehicles to rebroadcast alerts using a suitability index. Its hybrid V2V/V2I architecture shifts a significant portion of communication to less-congested RSU links, reducing contention and ensuring timely delivery of alerts.

These efficiencies also yield environmental benefits. As seen in Fig. 11 and 17, ICDDR leads to the lowest CO₂ emissions in both scenarios by minimizing idling and optimizing travel times. In Manhattan, CO₂ emissions drop by 34.9% compared to FCR, 20.7% versus RIDER, and 42.5% compared to PANDORA.

At 200 veh/km², average travel time in New Delhi reached 673 seconds, compared to 546 seconds in Manhattan—a 23.3% increase, due to more complex road structures. Yet ICDDR maintained its performance advantage across all metrics and models. This underscores the strength of its RSU-assisted architecture, capable of adapting to chaotic and irregular urban conditions.

V. CONCLUSION

This paper proposed a dynamic traffic control framework for the detection of congestion, the dissemination of alerts, and the rerouting of vehicles to avoid congested roads, to reduce the average travel time and the CO₂ emission of vehicles on high-density road networks. The proposed ICDDR framework dynamically detects congestion based on the average speed of the vehicles and the vehicle density of the road segments. It disseminate the congestion alert message using V2I and V2V communication modes and performs edge-assisted distributed processing, using Roadside Units (RSUs) as localized computational nodes for efficient rerouting decisions. The main goal of this proposed system is to reduce the travel time with very minimal messages transmitted. To incorporate this, the system disseminates messages through relevant vehicles, and thus, the network overhead is decreased. The RSUs are used as localized computational nodes for updating route weights based on real-time vehicle count and speed data. The framework decentralizes traffic processing, enabling dynamic modifications to route weights based on real-time vehicle density and speed data. To reroute the vehicles in the congestion zone, and the vehicles entering the congestion zone, our framework uses A* algorithm using the updated weights received from the RSUs and Euclidean distance as the heuristic, and it achieves enhanced rerouting, reduced travel time, reduced packet transmissions and packet collisions. This work can be extended by suggesting the optimal placement of RSUs and incorporating event and weather-driven congestion forecasting, and evaluating the system on live traffic of urban road networks.

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